CASCADIA ANALYTICS

AeroVironment

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Contract Momentum Obscured by Integration Strains and Dilution Overhang

Analyst: Justin Wen | October 29, 2026

AeroVironment presents an acute divergence between premier tactical franchise execution and severe corporate overhead pressures. While the core Autonomous Systems portfolio demonstrates robust programmatic demand evidenced by record funded backlog of $1.5B, the corporate structure remains burdened by significant acquisition digestion, accounting restatements, and negative free cash flow generation. At a market capitalization of $6.85B and reference spot price of $134.88, equity pricing embeds an aggressive recovery trajectory that fails to reflect balance sheet leverage and operational risks. We initiate coverage with a Hold rating and a probability-weighted blended price target of $82.04, representing -39.2% upside.

HOLD	$82.04	-39.2%	$134.88
Hold — Valuation Concerns	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE
INVESTMENT RATING			

The market remains transfixed by top-line expansion fueled by inorganic additions and high-profile contract awards, bidding shares near bull-case parameters. However, institutional capital must look past headline revenue to evaluate the underlying reality of goodwill impairments, working capital absorption, and negative cash flow that constrain equity value compounding across a multi-year horizon.

Executive summary

AeroVironment operates as a premier US defense company that makes military drones, loitering munitions, and related unmanned systems. The enterprise addresses critical operational requirements across modern conflict zones through specialized technological capabilities concentrated within its Autonomous Systems (AxS) and Space, Cyber, and Directed Energy (SCDE) segments. Across recent quarters, customer demand for tactical loitering munitions and small unmanned aircraft systems has accelerated, driving total funded backlog to a record level of $1.5B in Q1 FY2027 with a quarterly book-to-bill ratio of 1.4. This program visibility provides strong top-line support as modern military doctrines prioritize distributed, autonomous tactical strike platforms.

However, the operational achievements across the core unmanned systems business stand in stark contrast to the financial friction introduced by recent transformative scale. The acquisition and integration of BlueHalo have introduced pronounced organizational complexity, marked by severe contract volatility within the acquired business lines. Most visibly, a Space Force stop-work order on the SCAR/BADGER program during Q3 FY2026 catalyzed a $151.3M goodwill impairment and a $1.49B reduction to unfunded backlog. The friction compounded in Q4 FY2026 as the company restated its third-quarter financials to recognize an additional $89M impairment on the cancelled program, bringing full-year impairment charges to $241M and prompting management to disclose a material weakness in internal controls over financial reporting related to goodwill evaluations.

The enterprise financial print for FY2026 highlights the depth of this structural margin compression. Despite total revenue expanding by 140.9% to reach $1.98B, AeroVironment registered an operating loss of -$70.3M and a net loss of -$265.1M, translating to diluted earnings per share of -$5.40 and a return on equity of -6.0%. Free cash flow for FY2026 fell to -$164.6M, equivalent to a free cash flow margin of -8.3%, as operating profitability was weighed down by transaction amortization, integration expenses, and substantial working capital absorption. This cash burn continued into Q1 FY2027 with free cash flow of -$36M, while capital expenditures are projected to absorb 12-14% of revenue throughout FY2027.

Balance sheet leverage adds further constraints to capital allocation flexibility. AeroVironment concluded FY2026 with cash of $377.3M against total debt of $834.8M, resulting in a net debt burden of $457.5M. Furthermore, the transaction equity consideration drove 74% share dilution, expanding the diluted share count to 50.8 million shares. While Q1 FY2027 reflected operational stabilization with record revenue of $480.5M and adjusted EPS of $0.59 beating the $0.25 estimate, management reaffirmed full-year guidance calling for FY2027 adjusted EPS of $3.02-3.34, flat relative to $3.31 achieved in FY2026. Trading at 3.5x latest fiscal year revenue and 3.7x enterprise value to revenue, the market has detached from underlying fundamental returns on capital.

Operational snapshot

$134.88	$1.98B	$-164.6M	-8.3%	$457.5M	$134.88
REF. PRICE	FY2026 REVENUE	FY2026 FREE CASH FLOW	FCF MARGIN	NET DEBT	REF. CLOSE

Fiscal-year figures as reported in company filings and collected by Cascadia Analytics; ref. price as of 2026-10-29.

The moat is Specialized and Entrenched

AeroVironment maintains substantial structural advantages within the defense ecosystem, derived primarily from specialized technical know-how in unmanned systems, loitering munitions, and established programmatic integration with the US Department of Defense. Modern combat requirements increasingly rely on rapidly deployable, high-precision tactical strike capabilities where platforms such as Switchblade have established functional benchmarks. The scale of the company's customer entrenchment is underscored by programmatic vehicles such as the $990M indefinite delivery, indefinite quantity (IDIQ) contract, under which AeroVironment secured a $186M Switchblade task order in Q3 FY2026, creating high barriers to entry for competing platforms seeking sole-source status.

High switching costs further protect the company's core market share. Military procurement structures, extensive tactical training curricula, and established operational doctrines surrounding tactical drones ensure that military end-users face meaningful friction when transitioning across hardware architectures. The strength of this relationship is illustrated by the 21% year-over-year revenue growth in Autonomous Systems during Q1 FY2027 to reach $346M, alongside legacy business revenue expansion of 21% in Q2 FY2026 to $227.4M. This steady procurement cadence demonstrates that defense customers prioritize continuity of field-tested systems, preserving AeroVironment's volume base even during broader budget negotiations.

Furthermore, technical validation across small unmanned aircraft systems (UAS) reinforces pricing power and margin capture within the core hardware franchise. During Q4 FY2026, the company posted product gross margin of 44% supported by 31% organic volume growth, a doubling of Titan sales for the full year, and a $117M Low-Rate Initial Production award for the P550 platform. When insulated from service-heavy acquisition components, AeroVironment's proprietary intellectual property commands significant gross margin premiums over commoditized aerospace manufacturing peers.

Nevertheless, the structural moat is not completely impervious to contractual vulnerability. The swift termination of the SCAR/BADGER program by the Space Force in Q3 FY2026 illustrates that high switching costs at the tactical battlefield edge do not automatically extend across large-scale strategic space and cyber architectures. While the core tactical drone franchise remains highly protected, the SCDE segment remains subject to sudden budgetary reallocation,

preventing the enterprise from achieving the comprehensive defense prime insulation typical of legacy defense conglomerates.

Valuation

Three scenarios make the valuation range explicit

To establish a rigorous valuation range for AeroVironment, we utilize a multi-scenario discounted free cash flow framework spanning a forecast window from FY2027 to FY2036. The model evaluates cash flow compounding across 50.8 million diluted shares using an enterprise weighted average cost of capital of 10.16% and a 2.0% terminal growth rate in the base case. The enterprise baseline recognizes FY2026 revenue of $1.98B as the initial launch point, evaluating how top-line compounding and operating leverage drive long-term cash generation following the absorption of recent acquisition expenses.

Scenario parameters reflect distinctly divergent operational paths. In the Base Case, revenue compounds at a 12.0% CAGR through FY2036, supported by organic defense budget modernization, platform replacement cycles, and stabilization within acquired operations, while target free cash flow margins normalize to 12.0% as capital expenditure settles from current 12-14% levels. In the Bull Case, sustained international unmanned systems procurement and accelerated hardware adoption drive a 15.0% revenue CAGR, with operating scale lifting target free cash flow margins to 15.0%. Conversely, the Bear Case contemplates a deceleration to an 8.0% revenue CAGR paired with an 8.0% target free cash flow margin, reflecting structural margin impairment from fixed-price defense contracting hurdles and sustained integration friction.

Weighting these outcomes across a distribution of 25% bear, 50% base, and 25% bull yields a blended price target of $82.04 per share, indicating -39.2% downside from the reference spot price of $134.88. The valuation bridge indicates that while current equity trading levels approximate the bull-case target of $137.41, the broader distribution of operating outcomes points heavily toward downside re-rating, particularly given the near-term cash burn and structural margin constraints.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	8.0%	8.0%	$30.40	-77.5%
Base	50%	12.0%	12.0%	$80.18	-40.6%
Bull	25%	15.0%	15.0%	$137.41	+1.9%

Bear $30.40

Sustained integration friction, further program

Base $80.18

Autonomous Systems maintains double-digit

Bull $137.41

Rapid international scaling of loitering munitions,

cancellations within SCDE, and internal control remedies compress cash flows. Revenue compounds at only 8.0% while persistent capital expenditure requirements restrict terminal FCF margins to 8.0%, resulting in a target price of $30.40 (-77.5% vs spot).

growth while SCDE stabilizes to positive operating contributions. Revenue expands at a 12.0% CAGR through FY2036 with FCF margins normalizing to 12.0% as integration amortization rolls off, supporting a base target of $80.18 (-40.6% vs spot).

sustained Switchblade IDIQ delivery, and recovery across cyber programs accelerate revenue growth to 15.0% annually. Operating leverage expands FCF margins to 15.0%, producing an intrinsic equity value of $137.41 (+1.9% vs spot).

Valuation framework

A live discount rate, not a static hurdle

Our valuation framework implements an enterprise discount rate of 10.16%, derived from a capital asset pricing model that reflects the elevated operational and structural risks confronting AeroVironment. The capital structure has been fundamentally altered following transaction-related financing, leaving the firm with $834.8M in total debt against cash reserves of $377.3M, resulting in a net debt balance of $457.5M. The 10.16% hurdle rate properly incorporates this financial leverage alongside heightened equity beta driven by acquisition volatility, integration requirements, and defense budgetary uncertainties.

Current market pricing at $134.88 implies an investor return threshold substantially below 10.16%, effectively discounting an uninterrupted operational ramp that ignores documented friction points. By maintaining a dynamic 10.16% WACC across the FY2027 to FY2036 forecast window, our model penalizes the extended cash flow realization timeline necessitated by near-term negative free cash flow generation and sizable internal control compliance investments.

FCF margin normalization is the engine

The primary driver of intrinsic value in our framework is the trajectory of free cash flow margin normalization from historical deficits to terminal equilibrium. In FY2026, the company generated free cash flow of -$164.6M on $1.98B of revenue, translating to a negative free cash flow margin of -8.3%. This cash drain reflected severe working capital expansion, highlighted by unbilled receivables spiking $220M over a six-month span during the fiscal year, combined with transaction-driven amortization expenses totaling $48.2M in Q2 FY2026 alone.

Our model bridges this historical trough toward the scenario terminal targets of 8.0% in the bear case, 12.0% in the base case, and 15.0% in the bull case. Achieving the base-case 12.0% target requires management to successfully execute working capital rationalization, reverse the segment adjusted

EBITDA losses within SCDE (-$9M in Q1 FY2027), and moderate capital expenditures from the elevated 12-14% of revenue projected for FY2027 toward maintenance levels. Equity valuation cannot compound sustainably until operating cash flows decouple from working capital absorption.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

> **Methodology.** Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

KEY VALUATION PARAMETERS

Forecast window	**FY2027 to FY2036**
Live WACC	**10.16% (2026-10-29)**
Terminal growth	**2.0%**
Diluted shares	**50.8 million**

HOLD

over $82.04 blended target

Competitive moat and concentration risk

We rate the moat as Narrow and we view the market structure as Oligopoly (defense niches). We assign a premium of +12% to sector median multiples on moat strength alone.

- combat-proven Switchblade franchise plus funded backlog, LMS revenue +83% to $352m in FY2025, funded backlog $726.6m (nearly doubled YoY), and a US Army Switchblade award of up to $990m (FY2025 results coverage, 2025).
- FY2025 revenue $820.6m (+14% YoY), adjusted EBITDA $146.4m, FY2026 guidance $1.9-2.0bn revenue and $300-320m adjusted EBITDA following the $955m BlueHalo acquisition closed May 2025 (company results coverage, 2025). The guidance step-change is acquisition-led, so it evidences scale, not price increases, hence Narrow, not Wide.
- The weakest element is customer and conflict concentration, US Government was 74.7% of FY2025 revenue, international (largely Ukraine-driven) was 52.4%, and UxS revenue fell 15% to $381.8m in FY2025 as Ukraine sales eased. A backlog that doubles on a war is not the same as a moat that compounds in peacetime.

On a multiple cross-check, the justified range is 17.0-18.6x / 18.3-20.0x. The cross-check value is $80-$88 vs Base $68.69, spot $124.79, on a base of Framework base from the October 27 moat report. The gap verdict is In part (~35% of gap at top); market already pays more (spot P/E 28.4x).

Backlog is $1.5bn funded, about $2.8bn funded plus unfunded, 1 Aug 2026, with coverage of 9.1 months funded, 17.0 months total and visibility rated Medium on funded measure. On backlog alone, the stance is Supports guidance, not the model figure above guidance.

We score concentration and credit risk across seven families, where a High score is 8 points, a Medium score is 4 points, and a Low score is 1 point.

Risk family	Score
customer concentration	8 points
counterparty credit and funding dependence	1 points
single product, single program, or single platform dependency	4 points
single supplier, single fab, or single geography exposure	Not disclosed
platform, channel, or distribution dependency	1 points
customer insourcing risk	4 points
regulatory single point risk	8 points

The combined discount is 20 percent, capped at 20 percent, against an uncapped total of 26 percent. Shown side by side with the moat premium or discount, the net stance is -8 pp points arithmetically. We do not net these figures into a new target, and we do not double count a family excluded under the double-count rule.

The ladder ranks risk coarsely, and because 32 of the 47 names reach the 20 percent cap, the combined discount is not a mechanical valuation haircut until the ladder is recalibrated.

On the gap to spot, the concentration assessment reads as follows. Not at all

Both the moat premium or discount and the concentration discount are multiple and risk cross-checks dated October 27, 2026, and neither figure is in the DCF target itself. Cross-check figures were computed against the Base in the October 27 framework, so they may differ from a live Base that has since moved.

Business & segment drivers

Revenue prospects anchor on Autonomous Systems Expansion and Tactical Munitions Scaling HOLD

AeroVironment's operational performance remains segmented between its primary core franchise, Autonomous Systems, and the newly integrated Space, Cyber, and Directed Energy operations. The Autonomous Systems segment serves as the operational anchor, generating robust revenue growth across all recent reporting periods. In Q1 FY2027, Autonomous Systems revenue expanded by 21% year over year to reach $346M, demonstrating strong momentum following a pro forma revenue expansion of 25% in Q3 FY2026. This performance is sustained by the Department of Defense's accelerated adoption of loitering munitions and small unmanned systems across active combat theaters.

Conversely, the SCDE division has emerged as a primary operational drag on enterprise profitability. In Q1 FY2027, SCDE revenue declined by 21% year over year to $134.5M, generating a segment adjusted EBITDA loss of -$9M. This downturn follows the Space Force stop-work order on the SCAR/BADGER program disclosed in Q3 FY2026, which precipitated a $151.3M goodwill impairment and a direct $1.49B deduction from unfunded backlog, accompanied by a 19% pro forma revenue drop in that quarter. The persistent weakness within SCDE highlights the heightened execution risk inherent in integrating service-intensive, large-scale defense architectures relative to standardized hardware production.

The enterprise bookings trajectory underscores persistent structural demand for tactical platforms alongside near-term volatility. In Q1 FY2027, total company bookings stood at $683M, representing a 71% year-over-year increase and supporting a book-to-bill ratio of 1.4. This intake drove funded backlog to an all-time record of $1.5B, marking a 37% expansion year over year and a 23% sequential increase from the $1.2B reported at the end of FY2026. However, overall revenue visibility remains clouded by the full-year FY2026 GAAP net loss of -$265.1M and operating margin of -3.6%, which highlight the friction between securing sizable contract bookings and translating awards into GAAP profitability.

Management's forward guidance framework reflects this ongoing operational stabilization. For FY2027, management has reaffirmed top-line guidance of $2.125-2.225B and adjusted EBITDA of $305-325M, reflecting approximately 10% revenue expansion. However, forward adjusted EPS guidance of $3.02-3.34 remains flat relative to the $3.31 recorded in FY2026, underscoring that revenue growth is not generating immediate bottom-line expansion due to integration expenditures and elevated capital intensity.

What to monitor

- **Funded backlog trajectory** The company closed Q1 FY2027 with funded backlog at a record $1.5B, which serves as the critical indicator for conversion velocity into guided FY2027 revenue of $2.125-2.225B.
- **SCDE segment operating recovery** Monitoring whether the SCDE unit can reverse the -$9M adjusted EBITDA loss and 21% top-line decline reported in Q1 FY2027 following the cancellation of the SCAR program.
- **Cash flow conversion and working capital** Tracking whether free cash flow can emerge from negative territory following the -$36M burn in Q1 FY2027 and the substantial $220M unbilled receivables spike recorded during FY2026.
- **Internal control remediation** Assessing management's progress in resolving the material weakness in internal control over financial reporting associated with goodwill impairments following the $241M total write-downs in FY2026.

+140.9%	25.3%	-3.6%	-8.3%	$457.5M	-6.0%
FY2026 REVENUE GROWTH	GROSS MARGIN	OPERATING MARGIN	FCF MARGIN	NET DEBT	RETURN ON EQUITY

METRIC	FY2026 (AS REPORTED)
Revenue	$1.98B
Gross Profit	$500.6M
EBIT	$-70.3M
Net Income	$-265.1M
EBITDA	$-35.0M
Free Cash Flow	$-164.6M
Diluted EPS	$-5.40
ROE	-6.0%

Risk & moat matrix

The bear case

SHARE EROSION

Failure to maintain rapid platform innovation could allow peer defense contractors to contest key tactical UAS modernization programs. Any competitive loss within core unmanned systems would directly impair the revenue engine supporting enterprise operations.

M&A DIGESTION

Integrating BlueHalo has generated a $241M total goodwill impairment alongside material internal control weaknesses that continue to cloud reporting integrity. Protracted organizational friction risks further management distraction and additional segment write-downs.

REGULATORY AND PRICING SCRUTINY

Shifting priorities across federal defense appropriations can lead to sudden contract terminations, as demonstrated by the Space Force SCAR stop-work order. Such disruptions eliminate high-margin backlogs and leave the business exposed to

TECHNOLOGY DISRUPTION

Rapid technological obsolescence in counter-unmanned systems could degrade the battlefield utility of loitering munitions. Significant R&D allocations are continuously required to preserve electronic warfare resilience and mission success rates.

REINVESTMENT TREADMILL

Projected capital expenditures of 12-14% of revenue in FY2027 consume operational cash flow while the company absorbs elevated working capital burdens. This reinvestment intensity delays structural free cash flow normalization and restricts capital return flexibility.

WACC AND DURATION SENSITIVITY

At an enterprise discount rate of 10.16%, valuation multiples remain exceptionally sensitive to delayed cash conversion cycles. Sustained macroeconomic cost of capital pressures will penalize equity valuation if profitability metrics fail to materialize rapidly.

fixed overhead costs.

The strengths

OLIGOPOLY POSITION

AeroVironment holds an entrenched position as a premier provider of loitering munitions and tactical unmanned systems to the US military. Sole-source designations and extensive field deployment establish massive programmatic entry hurdles for prospective competitors.

FCF COMPOUNDING

The core Autonomous Systems business demonstrates robust operational leverage, achieving a 44% product gross margin in Q4 FY2026. Once acquisition amortization and working capital normalize, long-term annual free cash flow margins can scale toward target levels.

GEOGRAPHIC DURABILITY

Global demand for autonomous strike systems and tactical surveillance drones provides an enduring multi-year procurement backdrop. Modern warfare doctrine structurally prioritizes distributed, unmanned platforms over high-cost manned defense assets.

PIPELINE OPTIONALITY

A record funded backlog of $1.5B combined with a 1.4 book-to-bill ratio in Q1 FY2027 provides deep visibility into multi-year production volumes. Broad task order frameworks such as the $990M IDIQ provide continuous vehicle expansion opportunities.

The bottom line on risk and reward

The quantitative balance of risk and reward for AeroVironment remains unfavorable at prevailing valuation multiples. At a spot price of $134.88, the market is pricing equity shares near our bull-case target of $137.41, presuming near-flawless execution across both Autonomous Systems and troubled SCDE business lines. This pricing implicitly overlooks the structural reality that FY2027 adjusted EPS guidance of $3.02-3.34 is essentially flat relative to FY2026 levels of $3.31, while capital expenditure requirements remain elevated at 12-14% of total revenue.

Furthermore, the legacy of a -$265.1M net loss in FY2026, paired with -$164.6M in free cash flow and a net debt burden of $457.5M, demands a conservative valuation stance. The significant 74% share dilution executed to finance inorganic expansion has permanently expanded the share count to 50.8 million shares, reducing per-share upside while corporate integration continues to absorb working capital reserves. Until internal controls over goodwill impairments are definitively remediated and the business demonstrates sustained GAAP margin recovery, multiple compression remains the most probable outcome.

Our probability-weighted model computes a blended target of $82.04 per share, translating to -39.2% upside relative to current market trading levels. While the core drone franchise remains

exceptional, institutional investors should await a more favorable risk-adjusted valuation entry point. We maintain our Hold — Valuation Concerns rating on AeroVironment.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

Franchise deep dive

Autonomous Systems: Tactical Dominance Against Structural Headwinds

The foundational strength of AeroVironment resides within its core Autonomous Systems segment, which designs and manufactures military drones, loitering munitions, and related unmanned systems. The strategic value of this franchise has been validated by modern conflicts, where unmanned aerial systems have transitioned from niche reconnaissance tools into primary strike assets. The division's flagship products include the Switchblade loitering munition family, the P550 platform, and the Titan system, each serving specialized operational tiers within battlefield architectures.

Operational execution within Autonomous Systems has consistently outperformed broader corporate results. During Q4 FY2026, the segment delivered 31% organic growth and supported a company-wide product gross margin of 44%, alongside an adjusted EBITDA print of $140M that more than doubled prior-year performance. The momentum carried directly into Q1 FY2027, where Autonomous Systems revenue advanced 21% year over year to reach $346M. The durability of customer demand is substantiated by multi-year procurement programs, such as the $990M Switchblade IDIQ contract, which delivered a $186M task order during Q3 FY2026.

Platform diversification within Autonomous Systems provides further commercial support. The company secured a $117M Low-Rate Initial Production award for the P550 system in Q4 FY2026, while full-year sales of the Titan platform doubled across FY2026. Furthermore, total UAS operations demonstrated volume growth of 50% year over year during Q3 FY2026. This portfolio breadth protects AeroVironment from single-platform obsolescence, positioning the core franchise to capture sustained tactical budget allocations across international and domestic defense procurement cycles.

However, the exceptional performance of the Autonomous Systems division remains structurally burdened by the broader corporate footprint. The operating cash flow generated by tactical drone manufacturing is currently being absorbed to offset losses within the SCDE business and fund corporate capital expenditures. Until the enterprise can successfully ring-fence or remediate acquired operations, the premium cash flow profile of the core unmanned systems business will remain diluted by corporate-level integration strains.

What the recent record shows

- Autonomous Systems generated record revenue of $346M in Q1 FY2027, marking a 21% year-over-year increase and building on 25% pro forma growth in Q3 FY2026.

- Secured a major $186M Switchblade task order under an expansive $990M IDIQ contract during Q3 FY2026, reinforcing long-term program durability.
- P550 program reached a major commercialization milestone with a $117M Low-Rate Initial Production contract award announced in Q4 FY2026.
- Full-year sales of the Titan platform doubled during FY2026, supporting a 31% organic growth rate in Q4 FY2026 with a 44% product gross margin.

Our read: Autonomous Systems represents a world-class defense hardware franchise with durable competitive moats and exceptional customer entrenchment. However, with corporate valuation trading near bull-case limits and burdened by ongoing acquisition integration, the franchise's fundamental strength is already fully captured at current equity price levels.